

VISHNU VARDHAN AVASARALA

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EDUCATION

Bachelor of Engineering in Mechanical Engineering, Osmania University, India |

Aug'2018 - June'2022

TECHNICAL SKILLS

Programming Languages - Python, SQL ,Shell Scripting

Platforms - MLflow, CUDA, Kubeflow,Google Vertex AI, Microsoft Azure ML Studio, AWS SageMaker, OpenshiftAI

Libraries & Frameworks - NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, PyTorch, Transformers, OpenCV,JMeter, FastAPI, Flask

Databases & Data Stores - MySQL, PostgreSQL, MongoDB, Redis, PgVector

Cloud & DevOps - Amazon Web Services (AWS) (Lambda, EC2, S3, RDS, Bedrock), Docker, Kubernetes, Jenkins, CI/CD, ArgoCD, Terraform ,Azure Devops

Development & Collaboration Tools - Git, GitHub, Postman, Visual Studio Code, PyCharm, Spyder , Jupyter Notebook

Concepts & Methodologies -, Data Structures & Algorithms, Distributed Systems, Microservices, Machine Learning, Neural Networks, Agile/Scrum methodology

PROFESSIONAL EXPERIENCE

DevOps and Platform Engineer | *Tata Consultancy Services, Mumbai*

Sept'2023 - Mar'2026

- Architected end-to-end CI/CD pipelines using **Jenkins,Github Actions** and **ArgoCD**, automating **build,test,code and analysis**.
- Authored production-grade **Dockerfiles** for **ML model containerization**,leveraging optimized base images (**python:3.9-slim, TensorFlow/Pytorch**) and automated dependency management via **requirements.txt** --ensuring **100% environment consistency** across **dev, staging and production**.
- Packaged training model artifacts (**.pl.pth.h5**) with inference scripts into Docker containers, exposing model serving endpoints via FastAPI/Flask-reducing **model deployment setup by 70%** and enabling seamless **portabilty** in both **cloud** and **on prem environments**.
- Implemented Gitops-based deployment workflows with ArgoCD across **3 environments(dev, staging and production)**, achieving **100%** declarative infrastructure consistency and **reducing rollback time from hours to minutes under 5 minutes**.
- Extended **CI/CD and Gitops methodologies with MLOps** and unified entire process for continous training, continous integration and continous delivery.
- Operationalized ML model delivery with containerized packaging and **canary/blue green rollouts to inference endpoints, supporting zero-downtime**
- **model upgrades** with full versioning and auditability-reducing model rollback time by **70% across all the three environments**.

Cloud Engineer | *Urban Rebox IT Pvt , Hyderabad*

July'2022- Aug'2023

- Architected and orchestrated scalable, **multi-tenant** cloud infra across **Azure** and **AWS** supporting complex hybrid environments with **99.9% uptime**.
- Standardized infra delivery implementing **Iac tools like /Terraform,Bicep, ARM and AWS CDK**, reducing manual provisioning **time by 70%** and ensuring **auditable, reusable and version-controlled deployments** via **Git**.
- Unified hybrid-cloud management for global workloads by integrating **virtualized** environments(**Vmware,Nutanix,Proxmox VE**) with **Azure Arc**,achieving **centralized governance** and consistent policy enforcement.
- Strengthened organizational security by enforcing least privilege access through **Azure AD (Entra ID) and AWS IAM, Privileged Identity Management**
- Enforced **RBAC** to mitigate identity based risk, while securing sensitive application secrets, config files and certificates using **Azure Key Vault** .
- Spearheaded full-stack **observability** and **SIEM strategies** using **AWS CloudWatch, Azure Monitor** and **Splunk** and hardened enterprise cloud security posture by leveraging **Prisma Cloud for real time CSPM** and successfully remediating critical misconfigurations .
- Engineered a **multi-cloud cost optimization FinOps** framework that **reduced monthly spend by 60%** ; utilized **Savings Plans,Reserved Instances** and rigorous right sizing,Spot Instances,Auto Scaling,scheduled shutdown and implemented **tagging(Owner,Project,Environment) strategy** for cost allocation.

PROJECTS

NLP Sentiment analysis pipeline [Github](#)

Deployed and productionized a **DistilBERT**-based sentiment analysis model with a **FastAPI** inference layer achieving **sub-100ms real-time** prediction serving via containerized microservice achitecture.

Achieved **zero-touch model delivery** from code commit to live inference by building CI/CD triggers for automated model retraining in **multi-environments**.

Tools: Python, Openshift, Git, ArgoCD,GithubActions

Sentiment analysis API -Dockerised ML model

Built and containerized a production-ready Sentiment Analysis API by packaging the trained **NLP model** with a **REST** inference endpoint into a **lean 46.9MB Docker image**

Tools: NLP, Docker, FastAPI/Flask, Transformers

Certifications *Microsoft Certified: DevOps Engineer Expert, Microsoft Certified: Azure AI Engineer Associate, Microsoft Certified: Azure Data Scientist Associate, Microsoft Certified: Azure Developer Associate ,GitHub Foundations*